

Water extraction residue from maize milling by-product as a potential functional ingredient for the enrichment with fibre of cakes

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Highlights

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Maize milling by-product is a potential source of dietary fibre.

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Six cake formulations were developed (i.e. 2 particle sizes × 3 addition levels).

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Cake quality attributes were affected by powder particle size and addition level.

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Added by-product (up to 30%) induced great nutritional enhancement.

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Fibre-enriched formulations had satisfactory consumer acceptance.

Abstract

The present work is part of an effort for the holistic, environmental-friendly exploitation of maize milling by-product and emphasizes on the utilization of the rich in dietary fibre residue that remains after the aqueous extraction of oil for the preparation of health-promoting food products (e.g. bakery products, [breakfast cereals](#), dairy [desserts](#), sesame or peanut paste). To this direction, the residue was dehydrated and comminuted into two powders with different particle size that were then incorporated into the structure of yellow layer cakes at various wheat flour-substitution levels (up to 30%) and their impact on selected cake batter properties as well as the nutritional, textural, microstructural, thermal

and [sensory properties](#) and the ageing behaviour of the final products, were evaluated. Partial supplementation of the wheat flour by the dietary fibre-enriched powders had only a limited effect on some of the cakes' characteristics mainly induced by protein and starch dilution as well as limited [starch gelatinization](#), but, generally, their [sensory quality](#) was not negatively influenced, suggesting the use of both powders as a health-beneficial and inexpensive ingredient for the fortification of cakes with fibre.

Introduction

Formulation of fibre-rich bakery products has been constantly increasing over the last years in view of the existing link between diet and prevention of certain diseases (Fuller, Beck, Salman, & Tapsell, 2016; Veronese et al., 2018). The use of whole flour in baking or the incorporation in the product formula of traditionally undervalued by-products resulting from cereal dry milling or fruit processing, all rich sources of fibre constituents, have been tested as a means for the fortification of products such as breads, cakes and biscuits (Alba, Rizou, Paraskevopoulou, Campbell, & Kontogiorgos, 2020; Gómez, Manchón, Oliete, Ruiz, & Caballero, 2010a; Gómez, Moraleja, Oliete, Ruiz, & Caballero, 2010b; Lebesi & Tzia, 2012; Masoodi, Sharma, & Chauhan, 2002; Quiles et al., 2018; Sudha, Baskaran, & Leelavathi, 2007).

The by-product of dry corn milling is a material consisting mainly of the outer layers of grain and the germ and may easily become enriched in the latter by fractionation. This material finds use as animal feed or as raw material for the hexane extraction of corn oil. An alternative way to recover the corn oil from germ has been recently put forward (Nikiforidis & Kiosseoglou, 2009) based on the use of aqueous media in place of hexane. This much more environmental-friendly method leads to an extract consisting of intact oil bodies, well dispersed in a protein solution, which may find use in the preparation of emulsion-based food products, e.g. beverages or plant-based yogurts (Mantzouridou et al., 2019; Naziri et al., 2017). The water-extraction residue, after dehydration and milling, is turned into a powder that may then be fractionated by sieving into powders of various particle sizes, providing thus a series of very rich in fibre materials that may also be considered as potential functional ingredients for the enrichment with fibre of various food products.

In previous studies conducted by our team such powdered residues have been successfully tested to stabilize sesame seed paste against undesirable solid particle sedimentation and appearance of oiling-off phenomena upon storage or to enhance the nutritional properties of short-dough biscuits (Evlogimenou, Paraskevopoulou, & Kiosseoglou, 2017; Paraskevopoulou, Rizou, & Kiosseoglou, 2019). As a follow up of the above research work, the aim of this study was to investigate the potential of such powdered residues to serve as cheap fibre-rich additives for enhancing the nutritional

value of highly popular products, such as cakes. To this direction, wheat flour was partially replaced by the solid residue powder of two particle sizes and the physical, mechanical, microstructural, thermal, storage, nutritional and sensory properties of the cakes were studied. Taking into account the complexity of this type of products (considered as a dispersion of emulsified fat, air bubbles and flour particles in a continuous phase made up of egg white and yolk) in addition to the fact that they have more delicate flavour and textural characteristics compared to others (e.g. bread or biscuit), the degree to which their quality characteristics and consumer acceptability would be affected by the incorporation of the powders is considered of great importance.

Materials

Wheat flour (WF) ($12.2 \pm 0.2\%$ moisture; $10.8 \pm 0.2\%$ protein; $0.47 \pm 0.02\%$ ash; $1.82 \pm 0.4\%$ fat; $2.3 \pm 0.07\%$ total dietary fibre) was supplied by Chalkidiki Flour Mills S.A. (Chalkidiki, Greece). Margarine, granulated sugar, fresh whole eggs, fresh milk (3.5% fat) and baking powder were purchased from the local market. A rich in dry-milled maize germ (MG) industrial by-product, originating from maize (*Zea mays* L. ssp. *mays*) cultivated for the preparation of maize flour for baking, was kindly

Batter and cake physical properties

The effect of WF substitution by MGSR on cake batter density, baking loss and cross-sectional area is presented in Table 2. Results showed that MGSR incorporation brought about a significant decrease in the batter density only in the case of the 30% WF replacement in comparison to the control. A similar trend has also been reported by Gómez et al. (2010a) who attempted to evaluate the suitability of wholegrain flours from wheat, rye, barley etc. to produce yellow layer cakes of acceptable

Conclusions

Supplementation, at levels of up to 30% of the WF with the rich in dietary fibre powders, influenced to some extent the yellow-layer cake batter density and some of the final product characteristics, such as texture properties and staling behaviour, depending on the level of powder addition and particle size. TGA and SEM results suggested that partial hydration of starch and proteins may account for differences observed. The powder-containing products were equally well received as compared to

CRedit authorship contribution statement

Adamantini Paraskevopoulou: Conceptualization, Funding acquisition, Methodology, Formal analysis, Writing - original draft, Writing - review & editing, Visualization, Resources. **Ioanna Anagnostara:** Investigation, Visualization. **Georgia**

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Declaration of competing interest

None.

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